

203RR



Control board 230V for gate operators Instructions and warnings

ENGLISH

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OVERVIEW

These instructions were prepared by the manufacturer and are an integral part of the product. The operations described are designed for adequately trained and qualified personnel and must be carefully read and kept for future reference.



1 PRODUCT CONFORMITY

The 203RR programmable control board bears the EC label. DEA SYSTEM guarantees the conformity of the product to European Directives 2004/108/CE (concerning electromagnetic compatibility), 2006/95/CE (low voltage electrical equipment)



2 WARNINGS

Read these warnings carefully. Failure to respect the following warnings may result in dangerous situations.

-  **WARNING** DEA System reminds all users that the selection, positioning and installation of all materials and devices which make up the complete automation system, must comply with the European Directives 2006/42/CE (Machinery Directive), 2004/108/CE (electromagnetic compatibility), 2006/95/CE (low voltage electrical equipment). In order to ensure a suitable level of safety, besides complying with local regulations, it is advisable to comply also with the above mentioned Directives in all non-European countries. A1
-  **WARNING** Using the product under unusual conditions not foreseen by the manufacturer may cause dangerous situations; this is the reason why all the conditions prescribed in these instructions must be followed.
-  **WARNING** Under no circumstance must the product be used in an explosive environment or surroundings that may prove corrosive and damage parts of the product. A3

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- ⚠ **WARNING** To ensure an appropriate level of electrical safety always keep very low tension cables (controls, electric lock, aerial and auxiliary circuits power supply) apart from (min 4 mm if not insulated, 1 mm through insulation) 230V power supply cables. To keep them apart put cables inside a tubing tray and fasten them with clamps near the terminal board. Make sure cables are not damaged or worn off during installation. A4
- ⚠ **WARNING** Any installation, maintenance or repair operation on the whole system must be carried out exclusively by qualified personnel. All these operations must be performed only after disconnecting the power supply, and operating in strict compliance with the electrical standards and regulations in force locally. A5
- ⚠ **WARNING** See the user's manual of the operator by DEA System you are working on for further instructions on assembling and control board wiring operations (such as drilling holes to allow for wire passage, cable clamps use, etc). Failure to comply with these instructions may jeopardize the level of electrical safety. A6
- ⚠ **WARNING** During the motors stroke memorization, the control board detects automatically the presence and type of photocells, safety devices and limit switches which are installed. It is therefore essential that during this phase the latter be properly connected and working. A7
- ⚠ **WARNING** Wrong assessment of impact forces may cause serious damage to people, animal and things. DEA System reminds all personnel that the installer must ascertain that these impact forces, measured according to EN 12445 prescriptions, are actually below the limits indicated by EN12453 regulation. A8
- ⚠ **WARNING** Any external safety device installed in order to conform to the limits set for impact forces must comply with EN12978. A9
- ⚠ **WARNING** Using spare parts not indicated by DEA System and/or incorrect re-assembly may endanger people, animals and property, and may also cause malfunctioning of the product: always use parts provided by DEA System and follow assembly instructions A10
- ⚠ **WARNING** Disposal of packaging materials (such as plastic, card board, etc.) must be done according to regulations in force locally. Do not leave plastic bags and polystyrene within the reach of children



3 MODELS AND CONTENTS OF THE PACKAGE

The control board 203RR can be supplied together with DEA System gate operators for counterweight overhead doors, sliding doors and barriers, or they are supplied individually as spare part to be used only with DEA System above mentioned gate operators.



4 PRODUCT DESCRIPTION

203RR control board has been designed for the control of DEA System 230V gate operators with built-in magnetic encoder. It can therefore be used to automate counterweight overhead doors, sliding doors and barriers. It is extremely versatile, easy to install and fully complies with European regulations concerning electromagnetic compatibility and electric safety

Main features of the product:

1. setting all parameters by 3 keys and a 4-digit display;
2. possibility of fine tuning of motor speed both during its complete stroke and during the last phase of it (slow-down). It preserves motor torque even at very low speed;
3. encoder-controlled positioning system (only end of stroke or stop while closing are needed)
4. possibility to set at will the slow-down duration
5. Internal anti-crash safety device whose sensitivity can be adjusted (according to a 70-level scale) separately for both operating directions;
6. inputs to connect both normal and powered external safety devices (mechanical ribs or photocell barriers), with the possibility to run a self-test before each operation. Controlled photocells;
7. built-in 433,92MHz radio receiver for both HCS and HT12E coding offering the possibility to search and delete each transmitter separately.



- WARNING** DEA System reminds all users that the selection, positioning and installation of all materials and devices which make up the complete automation system, must comply with the European Directives 2006/42/CE (Machinery Directive), 89/336/CE and subsequent amendments (electromagnetic compatibility), 73/23/CE and subsequent amendments (low voltage electrical equipment). In order to ensure a suitable level of safety, besides complying with local regulations, it is advisable to comply also with the above mentioned Directives in all extra European countries.

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5 TECHNICAL DATA

Power supply	230 V ~ +/- 10% 50Hz
Flashing light output	230 V ~ max 40W art. Lumy
Auxiliary power supply output (24VAUX)	24 V ~ (max 200mA AUX+SIC)
Safety devices power supply output (24VSIC)	24 V ~ (max 200mA AUX+SIC)
Electric lock output	12 V ~ max 15VA (max 1 electric lock art. 115)
LC/SCA contact capacity	max 5A
Max motor capacity	2 X 500W max
Fuse F1	T5A 250V (retarded)
Fuse F2	T160mA 250V (retarded)
Radio receiver frequency	433,92 MHz coding rolling code / dipswitch
Max. number of transmitter controlled.....	100



6 OPERATING CONDITIONS

203RR control board is designed only to control DEA SYSTEM 230 V ~ gate and door operators, whose motor is equipped with magnetic encoder, and is therefore suitable for counterweight overhead doors, sliding doors and barriers. This control board has been designed and tested for operation under "normal" conditions for both residential and industrial use. The level of protection against dust and water is illustrated in the instructions booklet of DEA System gate operators equipped with built-in 203RR control boards.

- WARNING** Using the product under unusual conditions not foreseen by the manufacturer may cause dangerous situations; this is the reason why all the conditions prescribed in these instructions must be followed.
- WARNING** Under no circumstance must the product be used in an explosive environment or surroundings that may prove corrosive and damage parts of the product.

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7 ASSEMBLY AND WIRING INSTRUCTIONS

- WARNING** To ensure an appropriate level of electrical safety always keep very low tension cables (controls, electric lock, aerial and auxiliary circuits power supply) apart from (min 4 mm if not insulated, 1 mm through insulation) 230V power supply cables. To keep them apart put cables inside a tubing tray and fasten them with clamps near the terminal board. Make sure cables are not damaged or worn off during installation.
- WARNING** Any installation, maintenance or repair operation on the whole system must be carried out exclusively by qualified personnel. All these operations must be performed only after disconnecting the power supply, and operating in strict compliance with the electrical standards and regulations in force in the nation of installation.
- WARNING** See the user's manual of the operator by DEA System you are working on for further instructions on assembling and control board wiring operations (such as drilling holes to allow for wire passage, cable clamps use, ect). Failure to comply with these instructions may jeopardize the level of electrical safety.

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Connect to the power supply 230 V ~ $\pm$ 10% 50 Hz through a multi pole switch or a different device that can ensure multi pole disconnection from the power supply, with a contact opening of 3 mm. Use a cable with a minimum section of 3 x 1,5 mm² (e.g. a H07RN-F type).

Make all connections to the terminal board and remember to short-circuit, whenever necessary, all unused inputs. (See table 1 terminal board connection and Fig. 1 basic and complete wiring diagram)



Terminal board connection

1-2	Power supply input 230 V ~ +/-10% 50Hz
3-4	Free contact max. capacity 5A : this contact can be used to control an open gate warning light (P27=0) or a courtesy lamp (P27≠0)
5-6	Flashing light output 230 V ~ max 40W
7-8-9	Motor output max 2X 500W (7 opens, 8 common, 9 closes)
10-11	Electric lock output 12 V ~ 15VA
11	Common inputs
12	N.C. external safety device input. In case of activation it reverses the movement (P18=0) or it stops (P18=1). If unused, short circuit to terminal n°11
13	N.C. input end of stroke while opening. If unused, short circuit to terminal n°11
14	N.C. input end of stroke while closing. If unused, short circuit to terminal n°11
15	N.O. pedestrian opening button input. If activated, it opens partially of the gate
16	N.C. Photocell input. In case of activation it reverses the movement only while closing (P26=0) or it reverses the movement while closing and stops while opening (P26=1). If unused, short circuit to terminal n°11
17	N.C. stop input. If activated, it stops the movement during any operation If unused, short circuit to terminal n°11
18	N.O. open input. If activated, it opens or closes the gate. It can work in "reversal" mode (P25=0) or "step-by-step" mode (P25=1)
19	Aerial ground input
20	Aerial signal input
21-23	24 V ~ power supply output for auxiliary circuits and uncontrolled safety devices To be used as power supply for any auxiliary devices, photocell receivers (in all cases), and of safety devices when testing these latter before each gate operation
22-23	24 V ~ power supply output for controlled safety devices. To be used as power supply for photocell transmitters (in all cases) and of safety devices when testing these latter before each gate operation
23	Common safety devices

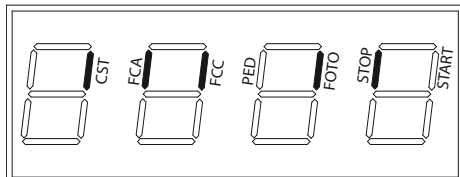


8 USE INSTRUCTIONS

After making all connections to the terminal board, remember to short-circuit, whenever needed, any unused input (see "connection to the control board") and power the card: on the display you will read for a few seconds "rES-" followed by the symbol "----" which stands for gate closed.

8.1 Visualization of inputs status

Press on the "OK" key to check that all inputs have been properly connected.



By pressing the "OK" key when the control board awaits further instructions ("- - -") the display shows some vertical segments: each one of them is associated to one of the control board inputs (see the picture above). When the segment is lighted it means that the contact associated to it is closed, on the contrary, when it is not lighted the contact is open. In order to do this:

8.2 Setup and memorization of the motor stroke

WARNING During motor stroke memorisation, the control board detects automatically the presence and type of photocells, safety devices and limit switches which are installed. It is therefore essential that during this phase the latter be properly connected and working.

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Instructions	Function	Display
	The control board is ready to receive instructions	----
Door/gate positioning		
	Scroll down the parameters until you visualize procedure P001	P001
	Confirm! The control board is ready for the positioning of the gate/door	P0-1
	Position gate/door in its standstill position after opening ⁽¹⁾	
	Confirm! The control board has memorized the gate/door position	P001
Motor stroke memorization		
	Scroll down the parameters until you visualize procedure P003	P003
	Confirm! The control board awaits a further confirmation	APP_r
	Confirm by pressing on the OK key for a few seconds! The procedure starts	APP_r
	Now the gate/door starts to close at a slow speed until it reaches the stroke end while closing (or the limit switch). ATTENZIONE: in case of operators without encoder (P034=1) and without limit switches, <u>at the arrival at the stroke press again the OK key, the operator stops.</u>	
	On the display you will read "----". Motor stroke memorization done!	----

⚠ WARNING If during the door opening you have no access to the push buttons of the control board, you may configure and memorize the stroke by using the buttons of a 4-channel remote control stored in memory

Instructions	Function	Display
	The control board is ready to receive instructions	----
Door/gate positioning and Motor stroke memorization		
	Scroll down the parameters until you visualize procedure P001	P001
	Confirm! The control board is ready for the positioning of the gate/door	P0-1
	Position gate/door in its standstill position after opening ⁽²⁾	
	Confirm! Now the gate/door starts to close at a slow speed until it reaches the stroke end while closing (or the limit switch).	APP_r
	On the display you will read "----". Motor stroke memorization done!	----

⁽¹⁾ By pressing on the key the gate/door must open, by pressing on the key the gate/door must close. If not, you must invert the two motor cables (terminals 7 and 9) and consequently the cables of limit switches too (if present).

⁽²⁾ Only if you use limit switches, first position the gate/door where you want it to stop in closing and then adjust the closing cam so that it presses on the limit switch associated to it in that point. Then position the gate/door in the opening position and adjust the opening cam so that it presses on the limit switches associated to it in that point.

8.3 Built-in radio receiver

DEA 203RR control board includes a 433,92MHz built-in radio receiver accepting both transmitters with HCS coding (complete rolling code or just fixed part), and HT12E dip-switch coding.

- The type of coding is selected by programming the working parameter n° 8 "type of coding" (see Table 2 Parameters)



- The receiver memory capacity can contain up to 100 different transmitters.
- When receiving a pulse from the transmitter, depending on your channel selection and linking, the start or the pedestrian inputs are activated. In fact, by programming one of the working parameters it is possible to choose, according to one's needs, which key of the memorized transmitters will activate the start input and which one will activate the pedestrian input (see "Channel selection and linking on the transmitter").
- While you memorize each transmitter the display shows a progressive number by which you will be able to trace and, if necessary, delete each transmitter individually

Instructions	Function	Display
	The control board is ready to receive instructions	----
Deletion of all transmitters		
	Scroll down the parameters until you visualize P004	P004
	Confirm! The control board awaits a further confirmation	CArC
	Confirm by pressing on the OK key for a few seconds! The procedure starts	CArC
	Done! The transmitters memory has been deleted	P004
	Scroll down the parameters until you visualize "----". The control board awaits a further confirmation	----
Memorization of transmitters ¹		
	Scroll down the parameters until you visualize P005	P005
	Confirm! The receiver enters in memorization mode The flashing light blinks!	LEAr
	Press on any key of the transmitter.	
	Memorization done! The flashing light goes out for 2 seconds The display visualizes the number of the transm. just memorized (es. "r001")	r001
	The receiver reverts automatically to memorization mode The flashing light blinks!	LEAr
	Memorize all necessary transmitters	
	Wait 10 seconds before quitting the memorization mode The receiver will now receive all the memorized transmitters	----
How to activate the memorization mode without operating on the control board ¹		
	Press simultaneously on key CH1 and CH2, or on the hidden key of a transmitter already memorized	LEAr
How to search and delete a transmitter		
	Scroll down the parameters until you visualize P006	P006
	Confirm! You can now select the transmitter	r001
	Scroll down the transmitter numbers until you reach the transmitter to be deleted (eg. "r003")	r003
	Confirm the deletion by pressing the OK key for a few seconds	r003
	OK! The transmitter is deleted	r----
	You can now select the parameter	P006
	Scroll down the parameters until you visualize "----". The control board awaits further instructions	----

¹ Make sure that the receiver is set to receive the type of coding of the transmitter you wish to memorize: visualize and, if necessary, update parameter n° 8 "Type of coding" (see "8.4 Personalization of working parameters")



Channel selection and linking on the transmitter

The built-in receiver can control both the start input and the pedestrian one. By setting the correct value of the parameter "P009 Selection and linking of radio channels" it is possible to decide which button of the transmitter will activate each input. If you check on the "working parameters" table you will realize that the P009 parameter allows you to choose among 16 different combinations. If, for instance, you attribute value "3" to the parameter P009, all memorized transmitters will activate the start input through the CH1 and the pedestrian input through CH4. Please see chapter "8.4 Personalization of working parameters" in order to select the right combination.

8.4 Personalization of working parameters

Instructions	Function	Display
	The control board is ready to receive instructions	----
	Scroll down the parameters until you visualize the one you wish to set (ex. P010)	P010
	Confirm! The display shows the set parameter value	d100
	Increase or decrease the value until you reach the value you wish to define	d080
	Confirm! The display shows again the parameter	P010
	Scroll down the parameters until you visualise "----". The control board awaits further instructions	----
The operator is now ready to work according to the new working parameters.		

8.5 Resetting of default parameters (p.007)

DEA 203RR control board software includes a reset procedure to restore default values (the one set by the maker) of all settable parameters, see Table 2 Parameters. The value originally set for each parameter is shown in the "working parameters table". In case you should reset all values and restore all default values, proceed as follows:

Instructions	Function	Display
	The control board is ready to receive instructions	----
	Scroll down the parameters until you visualize P007	P007
	Confirm! The control board awaits a further confirmation	dEF-
	Confirm by pressing on the OK button. The procedure starts	dEF-
	All parameters are now set at their original value	P007
	Scroll down the parameters until you visualise "----". The control board awaits further instructions	----

8.6 Safety devices

DEA 203RR control board allows installers to set up installations that truly comply with European regulations concerning garage doors and gates operators. More specifically, this control board allows you to comply with the limits set by the same regulations as to impact forces in case of collision with obstacles.

DEA 203RR control board is equipped with a built-in anti-crush safety device that, associated to the possibility of tuning up the motor speed, allows you to comply with the limits imposed by the above mentioned regulations in most installations.

In particular, you can adjust the anti-crush safety device sensitivity by properly setting the value assigned to the following parameters (see also "8.4 Personalization of working parameters"):

- P014 motor force in opening: from 30 (min. force, max sensitivity) to 100 (max force, neutralized sensitivity)
- P015 motor force in closing: from 30 (min. force, max sensitivity) to 100 (max force, neutralized sensitivity)

In case the gate structural features do not allow you to comply with the above force limits, it is possible to use external safety devices inputs (terminal no. 12). "CST" input can be configured by setting properly parameter no. 18:

- P018 = 0 "safety rib" mode functioning: when the input is activated, it reverses the movement direction
- P018 = 1 "photoelectric barrier" mode functioning: when the input is activated, it stops the movement



If the CST input is not used, shortcircuit it to terminal no.11. If you power the external safety devices by using 24V5IC output (terminal n° 22), their proper working is tested before each gate operation.

8.7 Messages shown on the display

203RR control board allows you to visualize on the display several messages concerning its working status and any malfunction:

Message	Description	
MESSAGES CONCERNING WORKING STATUS		
----	Gate is closed	
JL	Gate is open	
OPEN	Opening under way	
CLOS	Closing under way	
STEP	While in step-by-step mode, the control board awaits further instructions after a start command	
BLOC	Stop command received	
Barr	Cst activated while working in barrier mode	
ERROR MESSAGES		
Message	Description	Possible solutions
Err1 Err2	They point out that the gate has exceeded: - (Err1), the max allowed number of reversals (50) without ever reaching the end of stroke (or stop) while closing; - (Err2) the max number of uninterrupted operations (10) of the anti-crush safety device; therefore an “emergency operation” is under way: the control board sets the motors at a slow speed and searches the stops (or ends of stroke) in order to reset the positioning system. Once the stops (or ends of stroke) while closing are found again the message disappears and the control board awaits further instructions “----” and then resumes working normally.	In case the gate is not properly closed after the emergency operation (maybe because of false stops or obstacles due to mechanical frictions), proceed as follows: - Disconnect the power supply, check manually that no particular frictions and/or obstacles are present during the complete stroke of the gate/door. Leave the gate/door half-open. - Connect the power supply again and subsequently give a start pulse. At this point the gate/door will start to close at slow speed until reaching the stop (or end of stroke). Make sure that the gate operation is properly completed. Adjust force and motor speed values, if needed. If the gate keeps working improperly repeat the motor stroke memorization procedure (see paragraph 8.2)
Err3	External photocells and/or safety devices are activated or out of order	Make sure that all safety devices and/or photocells installed are working properly.
Err4	The motors are not connected or signals control board failure	Make sure that the motors are properly connected. If the message reappears change the control board.
Err6	Possible motor overheating due to obstacles hindering the gate/door movement. The control board does not respond to instructions	Remove any obstacle and wait until the message “Err6” is replaced by message “bLOC” and the control board responds to instructions again (a few seconds)
Err7	Operators movement not detected.	Make sure that operators and encoders connections are well done. Check the setting of parameter P034 (Motor selection with or without encoder) and make sure it is correct. If the error recurs, replace the encoder reader, and possibly the control panel.



9 MAINTENANCE

⚠ WARNING Any installation, maintenance or repair operation on the whole system must be carried out exclusively by qualified personnel. All these operations must be performed only after disconnecting the power supply, and operating in strict compliance with the electrical standards and regulations in force in the nation of installation.

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⚠ WARNING With control boards range "RR" disconnect the power supply wires before unlocking the operator manually. When you start the operator again the first operation will bring the door to a complete closing. If you do not follow this procedure the door will lose its right positioning.



10 PRODUCT DISPOSAL



In line with EU Directive 2002/96/EC for waste electrical and electronic equipment (WEEE), this electrical product must not be disposed of as unsorted municipal waste. Please dispose of this product by returning it to your local municipal collection point for recycling.



11 COMPLETE CLOSING ASSEMBLY

Remember that everyone who sells and/ or automates doors/gates becomes the manufacturer of the automated door/gate machine, and must therefore prepare and preserve a technical folder containing the following documents (see Machinery Directives Enclosure V).

- Assembly drawing of the automated door/gate.
 - Electrical connection and control circuit diagram.
 - Risk analysis including: a list of the essential safety requirements provided in machine Directive Enclosure I; a list of the risks posed by the door/gate and a description of the implemented solutions
- The installer must also:
- Keep these operating instructions and the instructions for all other components in a safe place.
 - Prepare the operating instructions and general safety warnings (by filling up these operating instructions) and hand a copy to the end user.
 - Fill in the maintenance handbook and hand a copy to the end user.
 - Draw up the EC declaration of conformity and hand a copy to the end user.
 - Fill in the complete EC label or plate and apply it to the automated door/gate.

N.B. The technical folder must be kept for inspection by the competent national authorities for at least ten years starting from the date of manufacturing of the door/gate operator.

⚠ WARNING DEA System reminds all users that the selection, positioning and installation of all materials and devices which make up the complete automation system, must comply with the European Directives 2006/42/CE (Machinery Directive), 2004/108/CE (electromagnetic compatibility), 2006/95/CE (low voltage electrical equipment). In order to ensure a suitable level of safety, besides complying with local regulations, it is advisable to comply also with the above mentioned Directives in all extra European countries.

⚠ WARNING Wrong assessment of impact forces may cause serious damage to people, animal and things. DEA System reminds all personnel that the installer must ascertain that these impact forces, measured according to EN 12445 prescriptions, are actually below the limits indicated by EN12453 regulation.

⚠ WARNING Any external safety device installed in order to conform to the limits set for impact forces must comply with EN12978.



PROCEDURE		PROCEDURE DESCRIPTION
P001	Positioning of gate/door	
P002	Unused parameter	
P003	Memorization of the motor stroke	
P004	Deletion of the radio receiver memory	
P005	Memorization of transmitters	
P006	Search and deletion of a transmitter	
P007	Resetting of default parameters	
PARAMETER DESCRIPTION		PARAMETER DESCRIPTION
P008	Type of coding of the radio receiver	
P009	Channel selection and linking to "start" and "pedestrian" inputs	
P010	Motor speed during normal stroke (calculated as % of max speed)	
P011	Motor speed during slow-down phase (calculated as % of max speed)	
P012	Slow-down duration (expressed as % of total stroke)	
P013	Closing acceleration length, from the starting at completely opened gate:	
P014	Motor force in opening (if = 100 --> max force, obstacle impact sensibility deactivated)	
P015	Motor force in closing (if = 100 --> max force, obstacle impact sensibility deactivated)	
P016	margin adjustment of the ground stop: adjusts the duration of the last stretch of the run (while opening and while closing) during which any obstacle is interpreted as a ground stop, stopping the engine without making an inversion.	

SETTABLE VALUES ¹					USER ²
000	HCS fixed part only				
001	HCS rolling code				
002	HT12E dip switch				
	START	PEDE- STRIAN		START	PEDE- STRIAN
001	CH1	CH2	009	CH3	CH4
002	CH1	CH3	010	CH4	CH1
003	CH1	CH4	011	CH4	CH2
004	CH2	CH1	012	CH4	CH3
005	CH2	CH3	013	CH1	CH2 ³
006	CH2	CH4	014	CH2	CH2 ³
007	CH3	CH1	015	CH3	CH2 ³
008	CH3	CH2	016	CH4	CH2 ³
	50.....100				
15	50.....100				
10	25	50			
000	Acceleration = 1sec				
001	Acceleration = 1,2sec				
002	Acceleration = 1,5sec				
003	Acceleration = 1,8sec				
	30	50,100			
	30	50,100			
1	9	30			



P017	Selection of the type of CST input: if=0 input for safety devices with N.C clean contact; if=1 input for safety devices with output at constant resistance	000	N.C. clean contact input	
P018	Selection of type of external safety device: rib / barrier. If the "rib" mode is selected, when the CST input is activated the movement direction is inverted; if the "barrier" mode is selected, when the CST input is activated the movement is stopped.	001	8K2 constant resistance input	
		000	safety ribs mode	
P019	Time of automatic closing (expressed in sec). If = 0 the automatic closing is deactivated	0 10	photoelectric barriers mode	
		0 10	255	
P020	Time of pre-flashing (expressed in sec)	0 2	deactivated	
P021	Recording delay of closing limit switch: if = 1, when the door arrives on FCC, the operator stops after about 1 sec. delay. If door arrives at the stroke while this delay, the operator immediately stops. Figure set by the factory: "0"	000	deactivated	
		001	activated	
P022	Search of the stop in opening too: when activated (P022>0) the operator stops only at the opening stop. Value settled indicates the delay (expressed in sec.) of the opening stop.	000	deactivated	
		>000	activated	
P023	If deactivated (P022=0) the operator stops in the point memorized during the learning. Collectivity function: if it is activated it deactivates both start and pedestrian inputs for the whole duration of automatic opening and closing	000	deactivated	
		001	activated	
P024	Ram blow function: if it is activated, it pushes the motors close for one second before each opening movement, so as to ease the releasing of any electric lock	000	deactivated	
		001	activated	
P025	Operating program: reversal (start->open, start->close, start->open ...), step-by-step (start->open, start->stop, start->close...)	000	inversion	
		001	step-by-step	
P026	PHOTO input functioning: if=0 photocells are activated while closing and at start when gate is closed; if=1 photocells are always activated; if=2 photocells are activated while closing only. PHOTO input activation, when activated, provokes the inversion (while closing), the stop (while opening) and prevent the starting (when gate is closed).	000	photocells are activated while closing and when gate is closed	
		001	photocells are always activated	
		002	photocells are activated at closing only	
P027	Clean contact operation: - If = 0, open gate fixed warning light, the contact is always closed when the gate is moving or opened, it opens again only when the closing movement is completed - If = 1, open gate intermittent light, the contact is slow while opening and fast while closing, always closed when the gate is opened, it opens again when the closing movement is completed - If > 1 courtesy light, the contact is closed during every movement, it opens again when the motor stops according to a pre-settable delay (expressed in sec)	000	open gate fixed warning light	
		001	open gate intermittent warning light	
		>001	courtesy light with settable delay-off	
P028	Short reversal at end of stroke: when the gate/door reaches the end of stroke, it reverses shortly the movement so as to "release" the mechanical stress due to the gate/door's pressure on the end of stroke itself.	000	deactivated	
		001	activated	
P029	Ramp rate duration - If =0 the motor starts immediately at the selected speed - If = 1 the motor speeds up progressively until it reaches the selected speed	000	deactivated	
		001	activated	



PED1	PED1	"PED" input functioning -If=0, it ensures the door or gate closing in any position, "AP" input works normally. -If=1 the "PED" input starts the closing while "AP" starts the opening. -If=2 the "PED" input (permanent command) starts the closing, the "AP" input (permanent command) starts the opening. The gate stops at release. -If>2 "PED" starts the pedestrian opening. The selected value indicates the duration of the pedestrian stroke (expressed as a % of the total stroke). The "AP" input normally works.	PED1	Central lock	
			PED1	Separate lock	
			PED2	Man present command	
			>PED2	Pedestrian	
PED1	PED1	Limitation of motor force when it reaches the end of stroke while closing - If = 0, Limitation is disabled (force level at the end of stroke is set automatically) - If different from 0, it indicates the force value (expressed as a % of the max value) the motor exerts at the end of stroke while closing.	PED1		
PED2	PED2	Reaction at detection of an obstacle while opening - If=0 the door inverses travelling direction - If different from 0 the door inverses travelling direction only for the set time (expressed in sec)	PED2		
PED3	PED3	Reaction at detection of an obstacle while closing - If=0 the door reverses travelling direction - If different from 0 the door reverses travelling direction only for the time set (expressed in sec)	PED3		
PED4	PED4	Operation with or without encoder -If=0 the encoder is installed: the speed adjustment and the anti-crushing device are available -If=1 the encoder is not installed: the speed adjustment and the anti-crushing device are not available.	PED4	encoder is installed	
			PED4	encoder is NOT installed	
¹ The default value set by manufacturer at the factory is written underlined. ² Column reserved to the installer to fill in with the automation personalised parameters ³ Inactive channel.					

Table 2 Parameters					
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Table 2 Parameters